

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407234
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Hatano; Shinji

Actuator

Abstract

This actuator includes a movable body provided with a magnet and a yoke, and a support body provided with a case and a coil that vibrates the movable body. The case includes a first case member having a first end plate portion opposing the movable body from a Z1 direction, and a second case member having a second end plate portion opposing the movable body from a Z2 direction. The first case member includes a first case first side plate portion extending toward the Z2 direction from a Y1 direction end of the first end plate portion, and a first case second side plate portion extending toward the Z2 direction from a Y2 direction end of the first end plate portion. The first case second side plate portion includes a first projection portion. The coil holder is press-fitted between the first projection portion and first case first side plate portion.

Inventors:	Hatano; Shinji (Nagano, JP)
Applicant:	NIDEC INSTRUMENTS CORPORATION (Nagano, JP)
Family ID:	86468758
Assignee:	NIDEC INSTRUMENTS CORPORATION (Nagano, JP)
Appl. No.:	18/324126
Filed:	May 25, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230387772 A1	Nov. 30, 2023

Foreign Application Priority Data

JP	2022-086491	May. 27, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: H02K33/18 (20060101)

U.S. Cl.:

CPC H02K33/18 (20130101);

Field of Classification Search

CPC: H02K (33/00); H02K (33/02); H02K (33/16); H02K (33/18); H02K (5/04); H02K (35/02)

USPC: 310/25; 310/15; 310/12.01; 310/81; 310/80; 310/321; 310/20; 310/21; 310/28-30; 310/36-37; 310/40MM

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9774236	12/2016	Xu	N/A	H02K 33/04
9948170	12/2017	Jun	N/A	H02K 33/00
10003746	12/2017	Minamisawa	N/A	G03B 5/00
10315221	12/2018	Mao	N/A	B06B 1/045
10447130	12/2018	Mao	N/A	B06B 1/045
10447131	12/2018	Mao	N/A	H02K 33/16
10447134	12/2018	Mao	N/A	H02K 33/02
10596594	12/2019	Ling	N/A	H02K 33/02
10674278	12/2019	Zhou	N/A	H04R 9/025
10951104	12/2020	Takeda	N/A	H02K 41/031
11201531	12/2020	Jin	N/A	H02K 33/16
11716003	12/2022	Miesner	310/27	H02K 33/16
11909273	12/2023	Hatano	N/A	H01F 7/081
2009/0267423	12/2008	Kajiwara	310/38	H02K 33/02
2012/0170792	12/2011	Li	381/412	H04R 9/066
2013/0313919	12/2012	Nakamura	310/25	H02K 33/00
2016/0372998	12/2015	Xu	N/A	H02K 33/16
2017/0033627	12/2016	Xu	N/A	H02K 1/34
2017/0144191	12/2016	Mao	N/A	B06B 1/045
2017/0288525	12/2016	Lee	N/A	G03B 3/10
2017/0310203	12/2016	Takeda	N/A	H02K 33/16
2017/0373578	12/2016	Wu	N/A	H02K 33/16
2018/0059794	12/2017	Nakamura	N/A	H02K 33/02
2018/0248458	12/2017	Amemiya	N/A	H02K 55/04
2018/0297061	12/2017	Mao	N/A	H02K 5/04
2018/0297070	12/2017	Xu	N/A	B06B 1/045
2018/0297071	12/2017	Xu	N/A	B06B 1/045
2018/0297077	12/2017	Chai	N/A	B06B 1/045
2018/0351442	12/2017	Liu	N/A	H02K 33/16
2019/0081544	12/2018	Zhu	N/A	H02K 33/02
2019/0190365	12/2018	Takeda	N/A	H02K 33/18
2019/0305630	12/2018	Mori	N/A	H01F 27/2823
2019/0305658	12/2018	Mori	N/A	H02K 33/12

2019/0305660	12/2018	Tsuchihashi	N/A	H02K 33/16
2019/0315070	12/2018	Yanagisawa	N/A	H02K 33/16
2020/0044526	12/2019	Tang	N/A	H02K 33/18
2020/0044541	12/2019	Takeda	N/A	H02K 33/16
2020/0059147	12/2019	Kitahara	N/A	H02K 33/16
2020/0107134	12/2019	Mori	N/A	H04R 7/18
2020/0161955	12/2019	Kitahara	N/A	H02K 33/16
2020/0204054	12/2019	Mori	N/A	H02K 33/02
2021/0075306	12/2020	Little	N/A	H02K 33/02
2021/0359583	12/2020	Toyota	N/A	H02K 33/16
2022/0200428	12/2021	Shao	N/A	H02K 33/02
2022/0200433	12/2021	Mao	N/A	H02K 33/18
2022/0209634	12/2021	Ma	N/A	H02K 33/02
2022/0247295	12/2021	Takei	N/A	H02K 9/223
2022/0352801	12/2021	Kitahara	N/A	H02K 33/02
2022/0368206	12/2021	Ma	N/A	H02K 33/02
2023/0387771	12/2022	Hatano	N/A	H02K 1/34

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
6194288	12/2016	JP	N/A
2020102901	12/2019	JP	N/A
2017002950	12/2016	WO	N/A

OTHER PUBLICATIONS

“Search Report of Europe Counterpart Application”, issued on Oct. 20, 2023, pp. 1-10. cited by applicant

Primary Examiner: Almawri; Maged M

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Japan application serial no. 2022-086491, filed on May 27, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Field of the Invention

(2) At least an embodiment of the present invention relates to an actuator that vibrates a movable body.

Description of the Related Documents

(3) Japanese Patent Application Laid-Open No. 2020-102901 (JPA 2020-102901) discloses an actuator which is provided with a movable body including magnets, and a support body including a coil, and vibrates the movable body with respect to the support body by applying a drive current to the coil. This type of actuator uses an elastic body or a viscoelastic body as a connecting body which connects the support body and the movable body. When the movable body is vibrated, a reaction force corresponding to the vibration of the movable body is applied to the support body

through the connecting body. As a result, a user who is touching the support body can feel the vibrations.

(4) In the actuator of JP2020-102901, the support body includes a metal case that defines the outer shape of the actuator, and a coil holder made of resin. The coil is an air core coil and is disposed in a coil placement hole in the coil holder. A metallic plate is attached to the coil holder to cover, from both sides, the coil and a plate portion provided with the coil placement hole. The movable body includes a first yoke that opposes the coil from one side, and a second yoke that opposes the coil from an other side, and magnets are fixed to each of the first yoke and the second yoke. The connecting body is disposed at both ends of the movable body in a longitudinal direction in locations where the yoke and the plate covering the coil oppose each other.

(5) The case of JP2020-102901 includes a first case member that covers the coil holder from one side, and a second case member that covers the coil holder from an other side. The coil holder includes side plate portions formed on an outer edge of the plate portion provided with the coil placement hole. The case and the coil holder are assembled by fitting the side plate portions of the coil holder inside the first case member and the second case member, which are manufactured by bending metal plates.

(6) In JP2020-102901, the first case member and the second case member are provided with engagement holes. The coil holder and the case are assembled by fitting engagement protrusion portions provided on the side surfaces of the coil holder into the engagement holes in each case member.

(7) Here, when assembling the coil holder inside the metal case, the dimensions are set such that a gap is formed between the case and the coil holder in order to ensure assembly even when there is a tolerance in the dimensions of the components. Therefore, the coil holder may move inside the case during vibration, which generates noise. In order to prevent this, a structure has been proposed in which an adhesive is inserted into the gap between the case and the coil holder such that they are fixed. However, the number of assembly steps may increase due to an additional injection step and curing step of the adhesive.

(8) Accordingly, at least an embodiment of the present invention is to suppress rattling of the coil holder against the case during vibration to reduce noise, and to suppress an increase in the number of assembly steps.

SUMMARY

(9) An actuator according to at least an embodiment of the present invention includes: a movable body; a support body provided with a metallic case that accommodates the movable body, and a coil holder made of resin; a connecting body that is connected to the movable body and the support body; and a magnetic drive circuit which is provided with a coil that is held by the coil holder, and a magnet that opposes the coil in a first direction, and which vibrates the movable body with respect to the support body in a second direction that intersects the first direction; wherein the case is provided with a first case member having a first end plate portion that opposes the movable body from one side in the first direction, and a second case member having a second end plate portion that opposes the movable body from an other side in the first direction, the first case member is provided with a first case first side plate portion that extends toward the other side in the first direction from an end on one side of the first end plate portion in a third direction, and a first case second side plate portion that extends toward the other side in the first direction from an end on an other side of the first end plate portion in the third direction, the third direction being a direction that intersects the first direction and intersects the second direction, the first case second side plate portion is provided with a first projection portion that protrudes toward the one side in the third direction, and the coil holder is press-fitted between the first projection portion and the first case first side plate portion.

(10) According to at least an embodiment of the present invention, the first projection portion is provided on the side plate portion (first case second side plate portion) on the other side of the first

case member in the third direction. Therefore, when the case and the coil holder are assembled, the first projection portion can be brought into elastic contact with the coil holder. Further, the coil holder can be positioned as a result of being biased toward the side plate portion (first case first side plate portion) on the one side in the third direction. Consequently, rattling of the coil holder inside the case can be suppressed, and the noise due to rattling of the coil holder can be suppressed. Furthermore, even when the dimensional accuracy of the first case member is low, the coil holder can be brought into contact with the first case first side plate portion with certainty, which provides good ease of assembly. In addition, because it is not necessary to fix the coil holder to the case using an adhesive, an injection step and curing step of the adhesive is not necessary. Therefore, an increase in the number of assembly steps can be suppressed.

(11) In at least an embodiment of the present invention, in the first projection portion, a surface on the other side in the third direction has a concave shape, and a surface on the one side in the third direction is a half punched portion having a convex shape. In this way, the first case member provided with the first projection portion can be easily manufactured. Therefore, an increase in component costs can be suppressed.

(12) In at least an embodiment of the present invention, the coil holder comprises a first holder member provided with a first coil holding portion disposed on the one side of the coil in the third direction, and a first holder side plate portion that extends in the first direction from an end of the first coil holding portion on the one side in the third direction, and a second holder member provided with a second coil holding portion disposed on the other side of the coil in the third direction, and a second holder side plate portion that extends in the first direction from an end of the second coil holding portion on the other side in the third direction, the first holder member and the second holder member are fixed via a metallic first plate, which overlaps the first coil holding portion and the second coil holding portion from the one side in the first direction, and a metallic second plate, which overlaps the first coil holding portion and the second coil holding portion from the other side in the first direction, the first projection portion makes contact with the second holder side plate portion from the other side in the third direction, and the first holder side plate portion makes contact with the first case first side plate portion from the other side in the third direction. As a result, by dividing the coil holder into two members by dividing it in the Y direction, a resin part that covers both sides of the coil in the X direction becomes unnecessary. Therefore, the external dimensions of the actuator in the X direction can be reduced. Alternatively, the coil can be made larger without increasing the size of the outer shape of the actuator, and the thrust of the magnetic drive circuit can be increased to generate large vibrations. Furthermore, because the coil can be protected by the first plate and the second plate, the possibility that the coil will be damaged due to collision with the magnet or collision with the yoke is low. In addition, the first holder member and the second holder member can be handled as the coil assembly as a result of being fixed via the first plate and the second plate. Therefore, even when the coil holder is divided into two members, a reduction in the ease of assembly of the actuator can be suppressed, and it is also possible for the first projection portion to be brought into elastic contact and positioned with respect to the coil holder.

(13) In at least an embodiment of the present invention, the coil holder is provided with a contact surface that makes contact with the first case first side plate portion from the other side in the third direction, and a substrate fixing portion that protrudes from the contact surface toward the one side of the first case first side plate portion in the third direction. In this way, a gap is formed in the third direction between the power feed substrate that is fixed to the substrate fixing portion, and the first case first side plate portion. Therefore, because contact between the metallic case and the power feed substrate can be avoided, it is possible to prevent the pattern on the power feed substrate and the case from making contact and short-circuiting.

(14) In at least an embodiment of the present invention, the first case member is provided with a first case third side plate portion that extends toward the other side in the first direction from an end

of the first end plate portion on one side in the second direction, and a first case fourth side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on an other side in the second direction, the second case member is provided with a second case third side plate portion that extends toward the one side in the first direction from an end of the second end plate portion on the one side in the second direction, and a second case fourth side plate portion that extends toward the one side in the first direction from an end of the second end plate portion on the other side in the second direction, and the first case third side plate portion and the first case fourth side plate portion are press-fitted between the second case third side plate portion and the second case fourth side plate portion. In this way, because the first case member and the second case member can be positioned by press-fitting, they do not need to be fixed using an adhesive. Therefore, an increase in the number of assembly steps can be suppressed.

(15) In at least an embodiment of the present invention, one of the second case third side plate portion and the first case third side plate portion is provided with a second projection portion that protrudes toward the other of the second case third side plate portion and the first case third side plate portion, and one of the second case fourth side plate portion and the first case fourth side plate portion is provided with a third projection portion that protrudes toward the other of the second case fourth side plate portion and the first case fourth side plate portion. In this way, a press-fitted state can be formed by bringing the leading edge of the second projection portion into elastic contact with the first case third side plate portion, and the leading edge of the third projection portion into elastic contact with the first case fourth side plate portion. Therefore, press-fitting can be performed with certainty even when the dimensional accuracy of the first case member and the second case member is low, and a reduction in the ease of assembly can also be suppressed.

(16) In at least an embodiment of the present invention, the second projection portion and the third projection portion are half punched portions. In this way, the second case member **40** provided with the second projection portion **46** and the third projection portion **47** can be easily manufactured because they can be easily formed by a method such as press working. Therefore, an increase in component costs can be suppressed.

(17) In at least an embodiment of the present invention, the movable body is provided with a yoke that holds the magnet, the yoke is provided with a first opposing portion that opposes the coil from the one side in the first direction, a second opposing portion that opposes the coil from the other side in the first direction, and a pair of connection portions that extend in the first direction on both sides of the coil in the second direction and connect the first opposing portion and the second opposing portion, the coil holder is provided with a pair of first holder projection portions that protrude toward the other side in the third direction from edges on both sides of the first holder side plate portion in the second direction, and a pair of second holder projection portions that protrude toward the one side in the third direction from edges on both sides of the second holder side plate portion in the second direction, an end portion of the first opposing portion and the second opposing portion on the one side in the third direction is disposed between the pair of first holder projection portions, and an end portion of the first opposing portion and the second opposing portion on the other side in the third direction is disposed between the pair of second holder projection portions. In this way, the first holder projection portions and the second holder projection portions function as stoppers that restrict the movable range of the movable body in the second direction. Therefore, because stoppers can be provided on the coil holder instead of the case, it is less likely that the movable body will collide with the case and be deformed or destroyed due to an impact caused by falling or the like, and the impact resistance of the actuator can be increased.

(18) In at least an embodiment of the present invention, the first case member is provided with a first case third side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on one side in the second direction, and a first case fourth side plate portion that extends toward the other side in the first direction from an end of the first end plate

portion on an other side in the second direction, and the pair of first holder projection portions and the pair of second holder projection portions are press-fitted between the first case third side plate portion and the first case fourth side plate portion. In this way, rattling of the coil holder **17** in the X direction can be suppressed. Furthermore, the first case third side plate portion and the first case fourth side plate portion are in a state where they are always pressed toward the outside by the coil holder. As a result, because the first case third side plate portion and the first case fourth side plate portion are always pressed against the second case third side plate portion and the second case fourth side plate portion from the inside, the first case member **30** can be positioned with respect to the second case member **40** in a state where it does not rattle. Moreover, because the parts that function as stoppers that regulate the movable range of the movable body **5** in the X direction can be utilized for press-fitted fixing, the press-fitted fixing can be firmly performed and the configuration of the coil holder **17** can be simplified.

(19) According to at least an embodiment of the present invention, the first projection portion is provided on the side plate portion (first case second side plate portion) on the other side of the first case member in the third direction. Therefore, when the case and the coil holder are assembled, the first projection portion can be brought into elastic contact with the coil holder. Further, the coil holder can be positioned as a result of being biased toward the side plate portion (first case first side plate portion) on the one side in the third direction. Consequently, rattling of the coil holder inside the case can be suppressed, and the noise due to rattling of the coil holder can be suppressed. Furthermore, even when the dimensional accuracy of the first case member is low, the coil holder can be brought into contact with the first case first side plate portion with certainty, which provides good ease of assembly. In addition, because it is not necessary to fix the coil holder to the case using an adhesive, an injection step and curing step of the adhesive is not necessary. Therefore, an increase in the number of assembly steps can be suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will now be described, by way of example only, with reference to the accompanying drawings which are meant to be exemplary, not limiting, and wherein like elements are numbered alike in several figures, in which:

(2) FIGS. **1A** and **1B** are perspective views of an actuator to which at least an embodiment of the present invention is applied when viewed from a Z2 direction and a Z1 direction.

(3) FIG. **2** is an exploded perspective view of an actuator showing a support body disassembled into a coil assembly, a first case member, and a second case member.

(4) FIG. **3** is a cross-sectional view of an actuator taken in the YZ plane (a cross-sectional view taken along position A-A in FIG. **1A**).

(5) FIG. **4** is a cross-sectional view of an actuator taken in the XZ plane (a cross-sectional view taken along position B-B in FIG. **1A**).

(6) FIG. **5** is an exploded perspective view of a movable body and a coil assembly when viewed from a Z2 direction.

(7) FIG. **6** is an exploded perspective view of a movable body and a coil assembly when viewed from a Z1 direction.

(8) FIG. **7** is an exploded perspective view of a coil assembly.

(9) FIG. **8** is a plan view of a coil holder and a coil.

(10) FIG. **9** is a cross-sectional view of an actuator taken in the XY plane (a cross-sectional view taken along position C-C in FIG. **3**) and partially enlarged views thereof.

(11) FIG. **10** is a cross-sectional view of an actuator showing the plan view shape and arrangement of a connecting body (a cross-sectional view taken along position D-D in FIG. **3**).

(12) FIG. 11 is a side view of a movable body, a connecting body, and a coil assembly when viewed from an X2 direction.

(13) FIG. 12 is a cross-sectional view of an actuator taken at the position of a press-fitted fixed portion (a cross-sectional view taken along position E-E in FIG. 9).

DETAILED DESCRIPTION

(14) An embodiment of an actuator to which at least an embodiment of the present invention is applied will now be described with reference to the drawings.

(15) Overall Configuration

(16) FIG. 1A is a perspective view of an actuator 1 to which at least an embodiment of the present invention is applied when viewed from a Z2 direction.

(17) FIG. 1B is a perspective view of the actuator 1 to which at least an embodiment of the present invention is applied when viewed from a Z1 direction. FIG. 2 is an exploded perspective view of the actuator 1, showing a support body 3 disassembled into a coil assembly 13, a first case member 30, and a second case member 40. FIG. 3 is a cross-sectional view of the actuator 1 taken in the YZ plane, and is a cross-sectional view taken along position A-A in FIG. 1A. FIG. 4 is a cross-sectional view of the actuator 1 taken in the XZ plane, and is a cross-sectional view taken along position B-B in FIG. 1A. FIG. 5 is an exploded perspective view of a movable body and the coil assembly 13 when viewed from a Z2 direction. FIG. 6 is an exploded perspective view of a movable body 5 and the coil assembly 13 when viewed from a Z1 direction. FIG. 7 is an exploded perspective view of the coil assembly 13. FIG. 8 is a plan view of a coil holder 17 and a coil 10. FIG. 9 is a cross-sectional view and partially enlarged views of the actuator 1 taken in the XY plane, and is a cross-sectional view taken along position C-C in FIG. 3.

(18) The actuator 1 is used as a tactile device that transmits information by vibrations. As illustrated in FIGS. 1A and 1B, the outer shape of the actuator 1 is substantially in the form of a rectangular parallelepiped. The actuator 1 generates vibrations in a transverse direction of the outer shape. In the description below, the transverse direction in which vibrations are generated will be referred to as the X direction (second direction). The longitudinal direction of the actuator 1, which is orthogonal to the X direction, will be referred to as the Y direction (third direction). The thickness direction (height direction) of the actuator 1, which is orthogonal to the X direction and the Y direction, will be referred to as the Z direction (first direction). Furthermore, the one side in the X direction will be referred to as the X1 direction, and the other side will be referred to as the X2 direction. The one side in the Y direction will be referred to as the Y1 direction, and the other side will be referred to as the Y2 direction. The one side in the Z direction will be referred to as the Z1 direction, and the other side will be referred to as the Z2 direction.

(19) As shown in FIGS. 1A to 4, the actuator 1 includes a support body 3 provided with a case 2 that defines the outer shape of the actuator 1, and a movable body 5 which is accommodated inside the case 2. Furthermore, the actuator 1 is provided with a connecting body 4 which connects the support body 3 and the movable body 5, and a magnetic drive circuit 6 which relatively moves the movable body 5 with respect to the support body 3 in the X direction (see FIGS. 3 and 4). As shown in FIG. 3, the connecting body 4 includes first connecting bodies 9A and second connecting bodies 9B.

(20) As shown in FIGS. 1A, 1B, and 2, the support body 3 is provided with a case 2 and a coil assembly 13. As shown in FIGS. 3 and 4, the coil assembly 13 includes a coil 10 disposed in the center of the actuator 1 in the Z direction. The case 2 is provided with a first case member 30 and a second case member 40 that are stacked in the Z direction. The first case member 30 is mounted on the coil assembly 13 from the Z1 direction. The second case member 40 is mounted on the coil assembly 13 and the first case member 30 from the Z2 direction.

(21) As shown in FIGS. 3 and 4, the movable body 5 includes a magnet 7 and a yoke 8. As shown in FIG. 4, the magnet 7 is fixed to an inner surface of the yoke 8, which is configured so as to surround the coil 10. The coil 10 and the magnet 7, which are opposed in the Z direction, constitute

the magnetic drive circuit **6**. The movable body **5** includes, as the magnet **7**, a first magnet **71** and a second magnet **72**. The first magnet **71** and the second magnet **72** are polarized into two poles in the X direction. When the movable body **5** and the support body **3** are assembled, the coil **10** opposes the first magnet **71** in the Z1 direction, and opposes the second magnet **72** in the Z2 direction.

(22) Coil Assembly

(23) As shown in FIG. 7, the coil assembly **13** is provided with the coil **10**, a first plate **11** stacked in the Z1 direction of the coil **10**, and a second plate **12** stacked in the Z2 direction of the coil **10**. The first plate **11** and the second plate **12** are made from a non-magnetic metal. Furthermore, the coil assembly **13** is provided with a first holder member **15** disposed on the Y1 side of the coil and a second holder member **16** disposed on the Y2 side of the coil **10**. The first holder member **15** and the second holder member **16** constitute a coil holder **17**. The first holder member **15** and the second holder member **16** are made from resin.

(24) As shown in FIGS. 2 and 3, the coil **10** is positioned at the center of the case **2** in the Z direction. The coil **10** is a flat air core coil, and the thickness direction thereof is oriented in the Z direction. As shown in FIG. 7, the coil **10** has an oval shape that is long in the Y direction, and includes a pair of long side portions **10a** and **10b** that extend parallel to each other in the Y direction. A central hole **10c** extending in the Y direction is provided between the pair of long side portions **10a** and **10b**. The long side portions **10a** and **10b** of the coil **10** oppose the first magnet **71** and the second magnet **72** in the Z direction.

(25) As shown in FIGS. 7 and 8, the first holder member **15** includes a first coil holding portion **151** disposed between the first plate **11** and the second plate **12**, a first holder side plate portion **152** that extends from the end of the first coil holding portion **151** on the Y1 side toward the Z1 direction and the Z2 direction, and a pair of first holder projection portions **153** that protrude in the Y2 direction from both ends of the first holder side plate portion **152** in the X direction. The first coil holding portion **151** is provided with recess portions **154** formed on both side surfaces in the X direction, and claw portions **155** that protrude from the center of the recess portions **154** in the Y direction. The pair of first holder projection portions **153** each extend to a position opposing the claw portions **155** in the X direction.

(26) A power feed substrate **14** is fixed to the first holder member **15**. In the present embodiment, the power feed substrate **14** is a rigid substrate. However, the power feed substrate **14** may be a flexible printed circuit board. Power is supplied to the coil **10** via the power feed substrate **14**. Two coil wires **10d** that have been drawn out from the coil **10** are disposed in groove portions **157** formed in the first coil holding portion **151**, drawn out to the Y1 side of the first holder side plate portion **152** from a notch portion **156**, in which a central part of the first holder side plate portion **152** in the X direction is notched toward the Z1 direction, and then bent toward the Z1 side and connected to the power feed substrate **14**.

(27) As shown in FIG. 7, the power feed substrate **14** is fixed to a substrate fixing portion **159** that protrudes from a contact surface **158**, which is a side surface of the first holder side plate portion **152** in the Y1 direction. The substrate fixing portion **159** surrounds the notch portion **156**, through which the coil wires **10d** are drawn out, on both sides in the X direction and on the Z1 side. As shown in FIG. 3, the end portion of the power feed substrate **14** on the Z1 side extends to a position further on the Z1 side than the end portion of the substrate fixing portion **159** on the Z1 side.

(28) As shown in FIGS. 7 and 8, the second holder member **16** includes a second coil holding portion **161** disposed between the first plate **11** and the second plate **12**, a second holder side plate portion **162** that extends from the end of the second coil holding portion **161** on the Y1 side toward the Z1 direction and the Z2 direction, and a pair of second holder projection portions **163** that protrude in the Y1 direction from both ends of the second holder side plate portion **162** in the X direction. The second coil holding portion **161** is provided with recess portions **164** formed on both side surfaces in the X direction, and claw portions **165** that protrude from the center of the recess

portions **164** in the Y direction. The pair of second holder projection portions **163** each extend to a position opposing the claw portions **165** in the X direction.

(29) The coil **10** is disposed between the first coil holding portion **151** and the second coil holding portion **161**. As shown in FIG. 7, the first plate **11** includes a plate-shaped first plate portion **111** that overlaps the coil **10** from the Z1 side. The first plate portion **111** includes a central plate portion **112** that overlaps the long side portions **10a** and **10b** of the coil **10**, a one side plate portion **113** that extends in the Y1 direction from the central plate portion **112** and overlaps the first coil holding portion **151**, and an other side plate portion **114** that extends in the Y2 direction from the central plate portion **112** and overlaps the second coil holding portion **161**. The width of the one side plate portion **113** and the other side plate portion **114** in the X direction is larger than the width of the central plate portion **112** in the X direction.

(30) A pair of central bent portions **115** that are bent in the Z2 direction are provided on both ends of the central plate portion **112** in the X direction. A pair of fixing bent portions **116** that are bent in the Z2 direction are provided on both ends of the one side plate portion **113** in the X direction and both ends of the other side plate portion **114** in the X direction. The fixing bent portions **116** in the four locations are each provided with a rectangular notch portion **117** that is notched toward the Z1 direction.

(31) The second plate **12** includes a flat second plate portion **121** that overlaps the coil **10** from the Z2 side. The second plate portion **121** includes a central plate portion **122** that overlaps the long side portions **10a** and **10b** of the coil **10**, a one side plate portion **123** that extends in the Y1 direction from the central plate portion **122** and overlaps the first coil holding portion **151**, and an other side plate portion **124** that extends in the Y2 direction from the central plate portion **122** and overlaps the second coil holding portion **161**. The width of the one side plate portion **123** and the other side plate portion **124** in the X direction is larger than the width of the central plate portion **122** in the X direction.

(32) A pair of central bent portions **125** that are bent in the Z1 direction are provided on both ends of the central plate portion **122** in the X direction. As shown in FIGS. 5 and 6, the leading edges of the central bent portions **125** oppose the leading edges of the central bent portions **115** of the first plate **11** in the Z direction. The central bent portions **115** of the first plate **11** and the central bent portions **125** of the second plate **12** cover the side surfaces of the long side portions **10a** and **10b** of the coil **10** in the X direction. A pair of fixing bent portions **126** that are bent in the Z1 direction are provided on both ends of the one side plate portion **123** in the X direction and both ends of the other side plate portion **124** in the X direction of the second plate **12**.

(33) When the coil assembly **13** is assembled, the fixing bent portions **116** in two locations on the Y1 side of the first plate **11** are fitted to the recess portions **154** of the first coil holding portion **151**, and the claw portions **155** are fitted to the notch portion **117** of each fixing bent portion **116** (see FIG. 9). As a result, the fixing bent portions **116** in two locations are locked to the side surfaces on both sides of the first coil holding portion **151** in the X direction.

(34) Similarly, the fixing bent portions **116** in two locations on the Y2 side are fitted to the recess portions **164** of the second coil holding portion **161**, and the claw portions **165** are fitted to the notch portion **117** of each fixing bent portion **116** (see FIG. 9). As a result, the fixing bent portions **116** in two locations are locked to the side surfaces on both sides of the second coil holding portion **161** in the X direction. As a result, the first holder member **15** and the second holder member **16** are positioned with respect to the first plate **11**.

(35) Next, the coil **10** is disposed between the first holder member **15** and the second holder member **16** assembled on the first plate **11**, and the central hole **10c** of the coil **10** is filled with an adhesive. Furthermore, the two coil wires **10d** that have been drawn out from the coil **10** are drawn out from the notch portion **156** of the first holder side plate portion **152** to the Y1 side. Then, the second plate **12** is placed thereon from the Z2 side and the adhesive is cured. As shown in FIGS. 3 and 4, an adhesive layer **18** of the cured resin is formed in the central hole **10c** of the coil **10**. As a

result, the second plate **12** is fixed to the coil **10** and the first plate **11** via the adhesive layer **18**. The adhesive filling the central hole **10c** enters the gap between the coil and the first coil holding portion **151** and the gap between the coil and the second coil holding portion **161** and hardens (not illustrated). This completes the coil assembly **13**.

(36) The width of the one side plate portion **123** and the other side plate portion **124** of the second plate **12** in the X direction is larger than that of the one side plate portion **113** and the other side plate portion **114** of the first plate **11**. Therefore, when the second plate **12** is placed on the coil **10**, the first coil holding portion **151**, and the second coil holding portion **161** from the Z2 side, as shown in FIGS. 5, 6, and 9, the first coil holding portion **151** is fitted between the fixing bent portions **126** in two locations on the Y1 side of the second plate **12**, and the second coil holding portion **161** is fitted between the fixing bent portions **126** in two locations on the Y2 side. Because the fixing bent portions **116** of the first plate **11** are fitted to the recess portions **154** of the first coil holding portion **151** and the recess portions **164** of the second coil holding portion **161**, they are covered from the outside by the fixing bent portions **126** of the second plate **12**.

(37) When the actuator **1** is manufactured, the coil **10**, the first plate **11**, the second plate **12**, the first holder member **15**, and the second holder member **16** are assembled as described above to complete the coil assembly **13**. Further, the yoke **8** and the magnet **7** are assembled so as to surround the coil assembly **13** to complete the movable body **5**, and the movable body **5** and the coil assembly **13** are connected by the connecting body **4**.

(38) Then, the first case member **30** and the second case member **40** are assembled with respect to the coil assembly **13** to accommodate the movable body **5** in the case **2**. The power feed substrate **14** may be attached to the coil assembly **13** before assembling the movable body **5**, or may be attached after assembling the case **2**.

(39) Yoke

(40) The yoke **8** is made of a magnetic material. As shown in FIGS. 3 and 4, the yoke **8** includes a first opposing portion **801** that opposes the coil **10** from the Z1 direction, and a second opposing portion **802** that opposes the coil **10** from the Z2 direction. The first magnet **71** is fixed to the first opposing portion **801**. The second magnet **72** is fixed to the second opposing portion **802**.

Furthermore, as shown in FIG. 4, the yoke **8** is provided with a pair of connection portions **803** that extend in the Z direction from both sides in the X direction of the coil **10**. The pair of connection portions **803** connects the first opposing portion **801** and the second opposing portion **802**.

(41) When the yoke **8** is assembled, a pair of second joining plate portions **805** that extends in the Z1 direction from both ends of the second opposing portion **802** in the X direction is press-fitted and fixed inside a pair of first joining plate portions **804** that extends in the Z2 direction from both ends of the first opposing portion **801** in the X direction. As a result, the pair of connection portions **803** is formed, and the yoke **8** is assembled in a shape that surrounds the outer peripheral side of the coil **10**, the first plate **11**, and the second plate **12** (see FIGS. 2 and 4).

(42) As shown in FIGS. 5 and 6, the yoke **8** includes a first yoke **81** and a second yoke **82**. The first yoke **81** is configured by two members that are joined together, namely a first inner member **83** that overlaps the coil **10** from the Z1 direction, and a first outer member **84** that overlaps the first inner member **83** from the Z1 direction. The second yoke **82** is configured by two members that are joined together, namely a second inner member **85** that overlaps the coil from the Z1 direction, and a second outer member **86** that overlaps the second inner member **85** from the Z1 direction.

(43) The first opposing portion **801** of the yoke **8** is configured by stacking the first outer member **84** and the first inner member **83** in the Z direction. As shown in FIGS. 5 and 6, the pair of first joining plate portions **804** is provided on the first outer member **84**, and extend toward the Z2 direction on both sides of the first inner member **83** and the first magnet **71** in the X direction. The first opposing portion **801** includes a pair of first connecting body fixing portions **806** that extend toward both sides of the first inner member **83** and the first magnet **71** in the Y direction. Each of the pair of first connecting body fixing portions **806** is connected to the first plate **11** via the first

connecting bodies **9A**.

(44) The second opposing portion **802** of the yoke **8** is configured by stacking the second outer member **86** and the second inner member **85** in the Z direction. As shown in FIGS. **5** and **6**, the pair of second joining plate portions **805** is provided on the second outer member **86**, and extend toward the Z1 direction on both sides of the second inner member **85** and the second magnet **72** in the X direction. The second opposing portion **802** includes a pair of second connecting body fixing portions **807** that extend toward both sides of the second inner member and the second magnet **72** in the Y direction. Each of the pair of second connecting body fixing portions **807** is connected to the second plate **12** via the second connecting bodies **9B**.

(45) Connecting Body

(46) As shown in FIGS. **3**, **5**, and **6**, the first connecting bodies **9A** and the second connecting bodies **9B** each have a rectangular parallelepiped shape that is long in the X direction. As shown in FIG. **3**, the first connecting bodies **9A** are positioned on the Z1 side of the coil **10**. The second connecting bodies **9B** are positioned on the Z2 side of the coil **10**. The first connecting bodies **9A** are disposed in two locations on the Y1 side and the Y2 side of the first magnet **71**, and are formed of two members having the same shape. The second connecting bodies **9B** are disposed in two locations on the Y1 side and the Y2 side of the second magnet **72**, and are formed of two members having the same shape. The first connecting bodies **9A** and the second connecting bodies **9B** each exhibit at least one of elasticity and viscoelasticity.

(47) As described above, the first connecting bodies **9A** are sandwiched between the first opposing portion **801** of the yoke **8** and the first plate **11** on both sides of the coil **10** in the Y direction. The first connecting bodies **9A** are compressed between the first opposing portion **801** and the first plate **11** in the Z direction. As described above, the second connecting bodies **9B** are sandwiched between the second opposing portion **802** of the yoke **8** and the second plate **12** on both sides of the coil **10** in the Y direction. The second connecting bodies **9B** are compressed between the second opposing portion **802** and the second plate **12** in the Z direction.

(48) In the present embodiment, the first connecting bodies **9A** and the second connecting bodies **9B** are gel-like members made from silicone gel. Silicone gel is a viscoelastic body whose spring constant when deformation occurs in an expanding/contracting direction is approximately three times greater than the spring constant when deformation occurs in a shear direction. When a viscoelastic body is deformed in the direction that intersects a thickness direction (shear direction), because this is a deformation in a direction in which the viscoelastic body is pulled and stretched, the viscoelastic body has deformation characteristics in which the linear component is larger than the nonlinear component. Furthermore, when a viscoelastic body is subjected to compression deformation by being pressed in the thickness direction, it has expansion/contraction characteristics in which the nonlinear component is larger than the linear component, but on the other hand, when the viscoelastic body is pulled and stretched in the thickness direction, it has expansion/contraction characteristics in which the linear component is larger than the nonlinear component.

(49) Alternatively, the first connecting bodies **9A** and the second connecting bodies **9B** may be formed using various rubber materials such as natural rubber, diene rubber (such as styrene-butadiene rubber, isoprene rubber, butadiene rubber, chloroprene rubber, and acrylonitrile-butadiene rubber), non-diene rubber (such as butyl rubber, ethylene-propylene rubber, ethylene-propylene-diene rubber, urethane rubber, silicone rubber, and fluoro-rubber), thermoplastic elastomers, and modified materials of these rubber materials.

(50) FIG. **10** is a cross-sectional view of the actuator **1** showing the plan view shape and arrangement of the connecting body **4**, and is a cross-sectional view taken along position D-D in FIG. **3**. As shown in FIG. **10**, the second connecting bodies **9B** are connected to the one side plate portion **123** and the other side plate portion **124** of the second plate **12**. As shown in FIG. the length of the second connecting bodies **9B** is larger than the width of the central plate portion **122** in the X direction, and larger than the width of the coil **10**, which is covered by the central plate portion **122**,

in the X direction.

(51) The first connecting bodies **9A** have the same shape as the second connecting bodies **9B**, and are disposed in the same positions as the second connecting bodies **9B** when viewed from the Z direction. The first connecting bodies **9A** are connected to the one side plate portion **113** and the other side plate portion **114** of the first plate **11**. The length of the first connecting bodies **9A** is larger than the width of the central plate portion **112** in the X direction, and larger than the width of the coil **10**, which is covered by the central plate portion **112**, in the X direction.

(52) Detailed Shape of Yoke

(53) FIG. **11** is a side view of the movable body **5**, the connecting body **4**, and the coil assembly **13** when viewed from the X2 direction. As described above, in the yoke **8**, both end parts of the first opposing portion **801** and the second opposing portion **802** in the Y direction protrude on the Y1 side and the Y2 side of the connection portions **803**. Further, the first connecting body fixing portions **806** and the first connecting bodies **9A** are disposed on the Z1 side of the connection portions **803**, and the second connecting body fixing portions **807** and the second connecting bodies **9B** are disposed on the Z2 side of the connection portions **803**. That is, as shown in FIGS. **10** and **11**, in the yoke **8**, both sides of the pair of connection portions **803** in the Y direction are notched in a shape that exposes the first connecting bodies **9A** and the second connecting bodies **9B** when viewed from the X direction. As shown in FIG. **10**, the first connecting bodies **9A** and the second connecting bodies **9B** oppose the first holder projection portions **153** of the first holder member **15** and the second holder projection portions **163** of the second holder member **16** in the X direction.

(54) When the movable body **5** is assembled around the coil assembly **13**, the connection portions **803** of the yoke **8** is disposed in the space between the leading edges of the first holder projection portions **153** in the Y2 direction and the leading edges of the second holder projection portions **163** in the Y1 direction (see FIGS. **2** and **9**). Here, each of the pair of connection portions **803** is provided with a first notch portion **808**, in which a central part in the Z direction of the edge on the Y1 side is notched toward the Y2 side, and a second notch portion **809**, in which a central part in the Z direction of the edge on the Y2 side is notched toward the Y1 side (see FIGS. **2** and **11**). As shown in FIG. **11**, when viewed from the X direction, the end portion of the first coil holding portion **151** on the Y2 side is disposed in the first notch portion **808**, and the end portion of the second coil holding portion **161** on the Y1 side is disposed in the second notch portion **809**. That is, in the yoke **8**, both end portions of the pair of connection portions **803** in the Y direction are notched in a shape that does not interfere with first coil holding portion **151** and the second coil holding portion **161**.

(55) Furthermore, when the movable body **5** is assembled around the coil assembly **13**, as shown in FIG. **2**, the first connecting body fixing portion **806** provided on the end portion of the first opposing portion **801** on the Y1 side and the second connecting body fixing portion **807** provided on the end portion of the second opposing portion **802** on the Y1 side are disposed between the pair of first holder projection portions **153** that are opposed in the X direction. In addition, the first connecting body fixing portion **806** provided on the end portion of the first opposing portion **801** on the Y2 side and the second connecting body fixing portion **807** provided on the end portion of the second opposing portion **802** on the Y2 side are disposed between the pair of second holder projection portions **163** that are opposed in the X direction. Therefore, the first holder projection portions **153** and the second holder projection portions **163** of the coil holder **17** collide with the first connecting body fixing portions **806** and the second connecting body fixing portions **807** and act as stoppers that regulate excessive movement of the movable body **5** when the movable body **5** moves significantly in the X direction due to an impact or the like.

(56) Case

(57) As shown in FIGS. **1A**, **1B**, and **2**, the first case member **30** includes a first end plate portion **31** having a substantially rectangular shape that opposes the coil **10** from the Z1 direction.

Furthermore, the first case member **30** includes a first case first side plate portion **32** that extends toward the Z2 direction from the end of the first end plate portion **31** in the Y1 direction, a first case second side plate portion **33** that extends toward the Z2 direction from the end of the first end plate portion **31** in the Y2 direction, a first case third side plate portion **34** that extends toward the Z2 direction from the end of the first end plate portion **31** in the X1 direction, and a first case fourth side plate portion **35** that extends toward the Z2 direction from the end of the first end plate portion **31** in the X2 direction.

(58) The edges of the first end plate portion **31** in the X direction are provided with projection parts in a central part in the Y direction that protrude on both sides in the X direction. As a result, the first case third side plate portion **34** and the first case fourth side plate portion **35** are each provided with a side plate central portion **301** that extends from the leading edge of the projection parts on the first end plate portion **31** toward the Z1 direction, and side plate end parts **302** that are provided in positions shifted further toward the inside (central side of the first case member **30** in the X direction) than the side plate central portion **301**, in both sides of the side plate central portion **301** in the Y direction. Both ends of the side plate central portion **301** in the Y direction are connected to a step portion that is substantially bent in a right angle toward the inside. The side plate central portion **301** and the side plate end parts **302** are connected via the step portions.

(59) As shown in FIG. 2, the first case first side plate portion **32** has a shape that does not interfere with the substrate fixing portion **159** provided on the first holder side plate portion **152**, and is notched in a shape that surrounds the Z1 side, the X1 side, and the X2 side of the substrate fixing portion **159**. The first case second side plate portion **33** is provided with circular first projection portions **36** that protrude toward the inside (Y1 direction) of the first case member **30**. The first projection portions **36** are half punched portions formed by deforming a metal plate into a shape that protrudes in the Y1 direction by press working. Therefore, the first projection portions **36** have a surface on the Y2 side with a concave shape that is recessed toward the Y1 side, and have a surface on the Y1 side with a convex shape that protrudes toward the Y1 side. The first projection portions **36** are formed in a leading edge part of the first case second side plate portion **33** in the Z2 direction in two locations that are separated in the X direction.

(60) In the present embodiment, when the first case member **30** is assembled with respect to the coil assembly **13**, the coil holder **17** is fitted between the first case second side plate portion **33** and the first case first side plate portion **32**. At this time, as shown in the partially enlarged view shown in FIG. 9, the first projection portions **36** provided in two locations on the first case second side plate portion **33** make contact with the second holder side plate portion **162**, which is disposed on the end portion of the coil holder **17** on the Y2 side. The first projection portions **36** make elastic contact with the first case second side plate portion **33** from the Y2 side and press the coil holder **17** to the Y1 side. As a result, the coil holder **17** is biased in the Y1 direction inside the first case member **30**, and the first holder side plate portion **152** provided on the end portion of the coil holder **17** on the Y1 side is pressed against the first case first side plate portion **32** from the Y2 side (see FIGS. 3 and 9). Consequently, the coil holder **17** is positioned with respect to the first case member **30** in a state where it does not rattle in the Y direction.

(61) The substrate fixing portion **159** provided on the coil holder **17** protrudes on the Y1 side from the notch provided in the first case first side plate portion **32** as described above. As shown in FIG. 3, in the present embodiment, the side surface of the first holder side plate portion **152** in the Y1 direction (contact surface **158**) is making contact with the first case first side plate portion **32**, and the step between the substrate fixing portion **159** and the contact surface **158** is larger than the plate thickness of the first case first side plate portion **32**. As a result, the end portion of the power feed substrate **14** on the Z1 side, which extends toward the Z1 side of the substrate fixing portion **159**, does not make contact with the first case first side plate portion **32**, and a space S (see FIG. 3) in the Y direction is formed between the power feed substrate **14** and the first case first side plate portion **32**. Therefore, it is possible to prevent the pattern on the power feed substrate **14** from

being short-circuited due to contact with the case 2.

(62) Next, the second case member **40** includes a second end plate portion **41** having a substantially rectangular shape that opposes the coil **10** from the Z2 direction. Furthermore, the second case member **40** includes a second case first side plate portion **42** that extends toward the Z1 direction from the end of the second end plate portion **41** in the Y1 direction, a second case second side plate portion **43** that extends toward the Z1 direction from the end of the second end plate portion **41** in the Y2 direction, a second case third side plate portion **44** that extends toward the Z1 direction from the end of the second end plate portion **41** in the X1 direction, and a second case fourth side plate portion **45** that extends toward the Z1 direction from the end of the second end plate portion **41** in the X2 direction.

(63) The edges of the second end plate portion **41** in the X direction are provided with projection parts in a central part in the Y direction that protrudes on both sides in the X direction. As a result, the second case third side plate portion **44** and the second case fourth side plate portion **45** are each provided with a side plate central portion **401** that extends from the leading edge of the projection parts on the second end plate portion **41** toward the Z2 direction, and side plate end parts **402** that are provided on both sides of the side plate central portion **401** in positions shifted further toward the inside (central side of the second case member **40** in the X direction) than the side plate central portion **401**, in both sides of the side plate central portion **401** in the Y direction. Both ends of the side plate central portion **401** in the Y direction are connected to a step portion that is substantially bent in a right angle toward the inside. The side plate central portion **401** and the side plate end parts **402** are connected via the step portions.

(64) The first case member **30** and the second case member **40** are positioned as a result of the pair of first side plate portions provided on both ends of the first case member **30** in the X direction (that is, the first case third side plate portion **34** and the first case fourth side plate portion **35**) being press-fitted between the pair of second side plate portions provided on both ends of the second case member **40** in the X direction (that is, the second case third side plate portion **44** and the second case fourth side plate portion **45**). After being temporarily fixed by press-fitting, the first case member **30** and the second case member **40** are fully fixed by welding the location where the second case third side plate portion **44** and the first case third side plate portion **34** overlap each other and the location where the second case fourth side plate portion **45** and the first case fourth side plate portion **35** overlap each other.

(65) As shown in FIG. 2, the second case third side plate portion **44** is provided with second projection portions **46** that protrude in the X2 direction (that is, toward the inside of the second case member **40**). Similarly, the second case fourth side plate portion **45** is provided with third projection portions **47** that protrude in the X1 direction (that is, toward the inside of the second case member **40**). The second projection portions **46** and the third projection portions **47** are formed in two locations on both ends of the side plate central portion **401** in the Y direction, and formed in one location on each of the side plate end parts **402** provided on both sides of the side plate central portion **401**. The second projection portions **46** and the third projection portions **47** are half punched portions having the same shape as the first projection portions **36**.

(66) The second projection portions **46** and the third projection portions **47** are formed in opposing positions in the X direction. The pair of first side plate portions (the first case third side plate portion **34** and the first case fourth side plate portion **35**) of the first case member **30** are press-fitted between the opposing second projection portions **46** and third projection portions **47**. As shown in the partially enlarged view of FIG. 9, the second projection portions **46** make elastic contact with the first case third side plate portion **34** from the X1 side. Furthermore, the third projection portions **47** make elastic contact with the first case fourth side plate portion **35** from the X2 side.

(67) As described below, in the present embodiment, because the coil holder **17** is press-fitted between the first case third side plate portion **34** and the first case fourth side plate portion a

pressing force in directions expanding toward both sides in the X direction always acts on the first case third side plate portion **34** and the first case fourth side plate portion **35**. Therefore, when the first case member **30** and the second case member **40** are assembled, a state is formed in which a pressing force that presses outward from the first case third side plate portion **34** and the first case fourth side plate portion **35** always acts on the second case third side plate portion **44** and the second case fourth side plate portion **45**.

(68) Press-Fitted Fixed Portion

(69) The coil holder **17** is fixed to the case **2** as a result of the first holder member **15** and the second holder member **16** each being press-fitted inside the first case member **30**. As shown in FIGS. **2** and **9**, the first holder member **15** includes a press-fitted fixed portion **50** that is press-fitted between the first case third side plate portion **34** and the first case fourth side plate portion **35**. Furthermore, the second holder member **16** includes a press-fitted fixed portion **60** that is press-fitted between the first case third side plate portion **34** and the first case fourth side plate portion **35**.

(70) FIG. **12** is a cross-sectional view of the actuator **1** taken at the position of the press-fitted fixed portion **50** (a cross-sectional view taken along position E-E in FIG. **9**). In the first holder member **15**, the first holder side plate portion **152** and the first holder projection portions **153** constitute the press-fitted fixed portion **50**. As shown in FIGS. **9** and **12**, the press-fitted fixed portion **50** is press-fitted between the pair of side plate end parts **302** provided on the end portion of the first case third side plate portion **34** and the first case fourth side plate portion **35** in the Y1 direction.

(71) As shown in FIGS. **9** and **12**, the end portion of the press-fitted fixed portion **50** in the X1 direction (that is, the first holder projection portion **153** on the X1 side) includes a one side fixing surface **51** that makes contact with the side plate end part **302** of the first case third side plate portion **34** from the inside. The end portion of the press-fitted fixed portion **50** in the X2 direction (that is, the first holder projection portion **153** on the X2 side) includes an other side fixing surface **52** that makes contact with the side plate end part **302** of the first case fourth side plate portion **35** from the inside.

(72) As shown in FIGS. **2**, **7** and **12**, the one side fixing surface **51** and the other side fixing surface **52** is provided on a central part of the press-fitted fixed portion **50** in the Z direction. The Z1 side and the Z2 side of the one side fixing surface **51** and the other side fixing surface **52** are each provided with step portions in the X direction. That is, the end portion of the press-fitted fixed portion **50** in the X1 direction is provided with a one side recess portion **53** on the Z1 side of the one side fixing surface **51** that is more recessed than the one side fixing surface **51** in the X2 direction, and a one side projection portion **54** on the Z2 side of the one side fixing surface **51** that protrudes further in the X1 direction than the one side fixing surface **51**. Similarly, the end portion of the press-fitted fixed portion **50** in the X2 direction is provided with an other side recess portion **55** on the Z1 side of the other side fixing surface **52** that is more recessed than the other side fixing surface **52** in the X1 direction, and an other side projection portion **56** on the Z2 side of the other side fixing surface **52** that protrudes further in the X2 direction than the other side fixing surface **52**.

(73) As shown in FIG. **12**, in the press-fitted fixed portion **50**, the one side fixing surface **51** and the other side fixing surface **52** provided in a central part in the Z direction make contact with the leading edge parts of the first case third side plate portion **34** and the first case fourth side plate portion **35** in the Z2 direction, and the parts in the Z1 direction (the one side recess portion **53** and the other side recess portion **55**) do not make contact with the first case third side plate portion **34** and the first case fourth side plate portion **35**. Therefore, the press-fitted fixed portion **50** presses the leading edge parts of the first case third side plate portion **34** and the first case fourth side plate portion **35**, which are in the Z2 direction, in directions expanding toward the outside (both sides in the X direction).

(74) The press-fitted fixed portion **50** is positioned in the Z direction with respect to the first case

member **30** as a result being press-fitted to a position in which the leading edges of the first case third side plate portion **34** and the first case fourth side plate portion **35** each make contact with a step portion between the one side projection portion **54** and the one side fixing surface **51** and a step portion between the other side projection portion **56** and the other side fixing surface **52**. The end portions of the press-fitted fixed portion **50** in the Z2 direction (the one side projection portion **54** and the other side projection portion **56**) are covered from the outside (both sides in the X direction) by the second case third side plate portion **44** and the second case fourth side plate portion **45** of the second case member **40**.

(75) Next, in the second holder member **16**, the second holder side plate portion **162** and the second holder projection portions **163** constitute the press-fitted fixed portion **60** (see FIGS. 2, 7 and 9). The press-fitted fixed portion **60** is press-fitted between the pair of side plate end parts **302** provided on the end portion of the first case third side plate portion **34** and the first case fourth side plate portion **35** in the Y2 direction.

(76) The press-fitted fixed portion **60** is configured in the same manner as the press-fitted fixed portion **50**. That is, the end portion of the press-fitted fixed portion **60** in the X1 direction includes a one side fixing surface **61** that makes contact with the side plate end part **302** of the first case third side plate portion **34** from the inside, a one side recess portion **63** on the Z1 side of the one side fixing surface **61** that is more recessed than the one side fixing surface **61** in the X2 direction (see FIG. 6), and a one side projection portion **64** on the Z2 side of the one side fixing surface **61** that protrudes further in the X1 direction than the one side fixing surface **61**. Furthermore, the end portion of the press-fitted fixed portion **60** in the X2 direction is provided with an other side fixing surface **62** that makes contact with the side plate end part **302** of the first case fourth side plate portion **35** from the inside, an other side recess portion **65** on the Z1 side of the other side fixing surface **62** that is more recessed than the other side fixing surface **62** in the X1 direction, and an other side projection portion **66** on the Z2 side of the other side fixing surface **62** that protrudes further in the X2 direction than the other side fixing surface **62**.

(77) In the press-fitted fixed portion **60**, the one side fixing surface **61** and the other side fixing surface **62** provided in a central part in the Z direction make contact with the leading edge parts of the first case third side plate portion **34** and the first case fourth side plate portion **35**, and the end portions in the Z1 direction (the one side recess portion **63** and the other side recess portion **65**) do not make contact with the first case third side plate portion **34** and the first case fourth side plate portion **35**. Therefore, the press-fitted fixed portion **60** presses the leading edge parts of the first case third side plate portion **34** and the first case fourth side plate portion **35**, which are in the Z2 direction, in directions expanding toward the outside (both sides in the X direction).

(78) Furthermore, the press-fitted fixed portion **60** is positioned in the Z direction with respect to the first case member **30** as a result being press-fitted to a position in which the leading edges of the first case third side plate portion **34** and the first case fourth side plate portion **35** each make contact with a step portion between the one side projection portion **64** and the one side fixing surface **61** and a step portion between the other side projection portion **66** and the other side fixing surface **62**. The end portions of the press-fitted fixed portion **60** in the Z2 direction (the one side projection portion **64** and the other side projection portion **66**) are covered from the outside (both sides in the X direction) by the second case third side plate portion **44** and the second case fourth side plate portion **45** of the second case member **40**.

(79) As a result of such a fixing structure, the pair of first side plate portions (the first case third side plate portion **34** and the first case fourth side plate portion **35**) of the first case member are always pressed toward the outside by a pressing force from the press-fitted fixed portions **50** and **60**. Therefore, a state is formed in which the first case third side plate portion **34** and the first case fourth side plate portion **35** are always pressed from the inside against the pair of second side plate portions (the second case third side plate portion **44** and the second case fourth side plate portion **45**) of the second case member **40** that has been placed on the first case member **30**. Therefore, the

first case member **30** is positioned with respect to the second case member **40** in a state where it does not rattle.

Main Effects of Present Embodiment

(80) As described above, the actuator **1** of the present embodiment includes: the movable body **5**; the support body **3** provided with the metallic case **2** that accommodates the movable body **5**, and the coil holder **17** made of resin; the connecting body **4** connected to the movable body **5** and the support body **3**; and the magnetic drive circuit **6** which includes the coil **10** held by coil holder **17** and the magnet **7** that opposes the coil **10** in the Z direction (first direction), and causes the movable body **5** to vibrate with respect to the support body **3** in the X direction (second direction). The case includes the first case member **30**, which is provided with the first end plate portion **31** that opposes the movable body **5** from the Z1 direction, and the second case member which is provided with the second end plate portion **41** that opposes the movable body **5** from the Z2 direction. The first case member **30** includes the first case first side plate portion **32** that extends toward the Z2 direction from the end of the first end plate portion **31** in the Y1 direction, and the first case second side plate portion **33** that extends toward the Z2 direction from the end of the first end plate portion **31** in the Y2 direction. The first case second side plate portion **33** is provided with the first projection portion **36** that protrudes in the Y1 direction, and the coil holder **17** is press-fitted between the first projection portion **36** and the first case first side plate portion **32**.

(81) In the present embodiment, one of the side plates of the first case member **30** that are opposed in the Y direction (first case second side plate portion **33**) is provided with a projection shape (first projection portion **36**). Therefore, when the case **2** and the coil holder **17** are assembled, the first projection portion **36** can be brought into elastic contact with the coil holder **17**. Further, the coil holder **17** can be positioned as a result of being biased toward the other side plate (first case first side plate portion **32**). As a result, rattling of the coil holder **17** inside the case can be suppressed, and the noise due to rattling of the coil holder **17** can be suppressed. Moreover, even when the dimensional accuracy of the first case member **30** is low, the coil holder **17** can be brought into contact with the first case first side plate portion **32** with certainty, which provides good ease of assembly. In addition, because it is not necessary to fix the coil holder **17** to the case **2** using an adhesive, an injection step and curing step of the adhesive is not necessary. Therefore, an increase in the number of assembly steps can be suppressed.

(82) In the present embodiment, in the first projection portion **36**, the surface in the Y2 direction has a concave shape, and the surface in the Y1 direction is a half punched portion having a convex shape. Therefore, the first case member **30** provided with the first projection portion **36** can be easily manufactured because it can be easily formed by a method such as press working. Therefore, an increase in component costs can be suppressed.

(83) In the present embodiment, the coil holder **17** is provided with: the first holder member which includes the first coil holding portion **151** disposed in the Y1 direction with respect to the coil **10**, and the first holder side plate portion **152** that extends in the Z1 direction and the Z2 direction from the end of the first coil holding portion **151** in the Y1 direction; and the second holder member **16**, which includes the second coil holding portion **161** disposed in the Y2 direction with respect to the coil **10**, and the second holder side plate portion **162** that extends in the Z1 direction and the Z2 direction from the end of the second coil holding portion **161** in the Y2 direction. The first holder member **15** and the second holder member **16** are fixed via the metallic first plate **11** that overlaps the first coil holding portion **151** and the second coil holding portion **161** from the Z1 direction, and the metallic second plate **12** that overlaps the first coil holding portion **151** and the second coil holding portion **161** from the Z2 direction. The first projection portion **36** makes contact with the second holder side plate portion **162** from the Y2 direction. The first holder side plate portion **152** is biased toward the Y1 direction inside the first case member **30** and makes contact with the first case first side plate portion **32** from the Y2 direction. As a result, by dividing the coil holder **17** into two members by dividing it in the Y direction, a resin part that covers both sides of the coil **10** in

the X direction becomes unnecessary. Therefore, the external dimensions of the actuator **1** in the X direction can be reduced.

(84) Alternatively, the coil **10** can be made larger without an increase in the outer shape of the actuator **1**, and the thrust of the magnetic drive circuit **6** can be increased to generate large vibrations. Furthermore, because the coil **10** can be protected by the first plate **11** and the second plate **12**, the possibility that the coil **10** will be damaged due to collision with the magnet **7** or collision with the yoke **8** is low. In addition, the first holder member **15** and the second holder member **16** can be handled as the coil assembly **13** as a result of being fixed via the first plate **11** and the second plate **12**. Therefore, even when the coil holder **17** is divided into two members, a reduction in the ease of assembly of the actuator **1** can be suppressed, and it is also possible for the first projection portion **36** to be brought into elastic contact and positioned with respect to the coil holder **17**.

(85) In the present embodiment, the coil holder **17** includes the contact surface **158** that makes contact with the first case first side plate portion **32** from the Y2 direction, and the substrate fixing portion **159** that protrudes from the contact surface **158** toward the Y1 direction of the first case first side plate portion **32**. As a result, a space S is formed in the Y direction (third direction) between the power feed substrate **14** that is fixed to the substrate fixing portion **159**, and the first case first side plate portion **32**. Therefore, because contact between the metallic case **2** and the power feed substrate **14** can be avoided, it is possible to prevent the pattern on the power feed substrate **14** and the case **2** from making contact and short-circuiting.

(86) In the present embodiment, the first case member **30** includes the first case third side plate portion **34** that extends toward the Z2 direction from the end of the first end plate portion **31** in the X1 direction, and the first case fourth side plate portion **35** that extends toward the Z2 direction from the end of the first end plate portion **31** in the X2 direction. The second case member **40** includes the second case third side plate portion **44** that extends toward the Z1 direction from the end of the second end plate portion **41** in the X1 direction, and the second case fourth side plate portion **45** that extends toward the Z1 direction from the end of the second end plate portion **41** in the X2 direction. The first case third side plate portion **34** and the first case fourth side plate portion **35** are press-fitted between the second case third side plate portion **44** and the second case fourth side plate portion **45**. In such a configuration where the first case member **30** and the second case member **40** can be positioned by press-fitting, they do not need to be fixed using an adhesive. Therefore, an increase in the number of assembly steps can be suppressed.

(87) In the present embodiment, the second case third side plate portion **44** is provided with the second projection portion **46** that protrudes toward the first case third side plate portion **34**, and the second case fourth side plate portion **45** is provided with the third projection portion **47** that protrudes toward the first case fourth side plate portion **35**. As a result, a press-fitted state can be formed by bringing the leading edge of the second projection portion **46** into elastic contact with the first case third side plate portion **34**, and bringing the leading edge of the third projection portion **47** into elastic contact with the first case fourth side plate portion **35**. Therefore, press-fitting can be performed with certainty even when the dimensional accuracy of the first case member **30** and the second case member **40** is low, and a reduction in the ease of assembly can also be suppressed.

(88) In the present embodiment, the second projection portion **46** and the third projection portion **47** are half punched portions. That is, in the second projection portion **46**, the surface in the X1 direction has a concave shape, and the surface in the X2 direction is a half punched portion having a convex shape. Furthermore, in the third projection portion **47**, the surface in the X2 direction has a concave shape, and the surface in the X1 direction is a half punched portion having a convex shape. Therefore, the second case member **40** provided with the second projection portion **46** and the third projection portion **47** can be easily manufactured because it can be easily formed by a method such as press working. Therefore, an increase in component costs can be suppressed.

(89) In the present embodiment, the movable body **5** is provided with the yoke **8** which holds the magnet **7**. The yoke **8** includes the first opposing portion **801** that opposes the coil **10** from the Z1 direction, the second opposing portion **802** that opposes the coil **10** from the Z2 direction, and the pair of connection portions **803** that extend in the Z direction on both sides of the coil **10** in the X direction and connect the first opposing portion **801** and the second opposing portion **802**. The coil holder **17** includes the pair of first holder projection portions **153** that protrude toward the Y2 direction from the edges on both sides of the first holder side plate portion **152** in the X direction, and the pair of second holder projection portions **163** that protrude toward the Y1 direction from the edges on both sides of the second holder side plate portion **162** in the X direction. The end portions of the first opposing portion **801** and the second opposing portion **802** in the Y1 direction are disposed between the pair of first holder projection portions **153**. The end portions of the first opposing portion **801** and the second opposing portion **802** in the Y2 direction are disposed between the pair of second holder projection portions **163**. As a result, the first holder projection portions **153** and the second holder projection portions **163** function as stoppers that regulate the movable range of the movable body **5** in the X direction. In this way, by providing stoppers on the coil holder **17** instead of the case **2**, it is less likely that the movable body **5** will collide with the case and be deformed or destroyed due to an impact caused by falling or the like. Therefore, the impact resistance of the actuator **1** can be increased.

(90) In the present embodiment, the first case member **30** includes the first case third side plate portion **34** that extends toward the Z2 direction from the end of the first end plate portion **31** in the X1 direction, and the first case fourth side plate portion **35** that extends toward the Z2 direction from the end of the first end plate portion **31** in the X2 direction. The pair of first holder projection portions **153** and the pair of second holder projection portions **163** are press-fitted between the first case third side plate portion **34** and the first case fourth side plate portion **35**. In this way, by press-fitting the coil holder **17** between the side plates (the first case third side plate portion **34** and the first case fourth side plate portion **35**) that are opposed in the X direction, rattling of the coil holder **17** in the X direction can be suppressed. Furthermore, the first case third side plate portion **34** and the first case fourth side plate portion **35** are in a state where they are always pressed toward the outside by the coil holder **17**. As a result, because the first case third side plate portion **34** and the first case fourth side plate portion **35** are always pressed against the second case third side plate portion **44** and the second case fourth side plate portion **45** from the inside, the first case member **30** can be positioned with respect to the second case member **40** in a state where it does not rattle. Moreover, because the parts that function as stoppers that regulate the movable range of the movable body **5** in the X direction can be utilized for press-fitted fixing, the press-fitted fixing can be firmly performed and the configuration of the coil holder **17** can be simplified.

Modifications

(91) (1) In the above embodiment, the first connecting bodies **9A** and the second connecting bodies **9B** are provided as the connecting body **4**. However, a configuration is possible in which only one of the first connecting bodies **9A** and the second connecting bodies **9B** is provided.

(92) (2) In the above embodiment, the first magnet **71** and the second magnet **72** are provided as the magnet **7**. However, a configuration is possible in which only one of the first magnet **71** and the second magnet **72** are provided.

(93) (3) In the above embodiment, in the yoke **8**, the first yoke **81** and the second yoke **82** are each configured stacking an inner member and an outer member. However, the first yoke **81** and the second yoke **82** may each be configured by only an outer member.

(94) (4) In the above embodiment, the yoke **8** has the pair of second joining plate portions **805** press-fitted inside the pair of first joining plate portion **804**. However, a configuration is also possible in which the pair of first joining plate portions **804** is press-fitted inside the pair of second joining plate portions **805**.

(95) (5) In the above embodiment, the coil holder **17** is split into two parts in the Y direction and is

configured by two members. However, the coil holder **17** may be configured by a single member. For example, the coil holder **17** can be configured having a coil placement hole provided in a plate portion connecting the first holder side plate portion **152** and the second holder side plate portion **162**.

(96) (6) The above embodiment has a configuration in which projection portions (the second projection portions **46** and the third projection portions **47**) that protrude toward the inside are provided on the side plates of the second case member **40** in the X direction (the second case third side plate portion **44** and the second case fourth side plate portion **45**). However, the projection portions may be provided on either of the first case member **30** and the second case member **40**. That is, the second projection portions that protrude toward the outside may be provided on the first case third side plate portion **34**, and the third projection portions that protrude toward the outside may be provided on the first case fourth side plate portion **35**.

Claims

1. An actuator comprising: a movable body; a support body provided with a metallic case that accommodates the movable body, and a coil holder made of resin; a connecting body that is connected to the movable body and the support body; and a magnetic drive circuit which is provided with a coil that is held by the coil holder, and a magnet that opposes the coil in a first direction, and which vibrates the movable body with respect to the support body in a second direction that intersects the first direction; wherein the case is provided with a first case member having a first end plate portion that opposes the movable body from one side in the first direction, and a second case member having a second end plate portion that opposes the movable body from an other side in the first direction, the first case member is provided with a first case first side plate portion that extends toward the other side in the first direction from an end on one side of the first end plate portion in a third direction, and a first case second side plate portion that extends toward the other side in the first direction from an end on an other side of the first end plate portion in the third direction, the third direction being a direction that intersects the first direction and intersects the second direction, the first case second side plate portion is provided with a first projection portion that protrudes toward the one side in the third direction, and the coil holder is press-fitted between the first projection portion and the first case first side plate portion.
2. The actuator according to claim 1, wherein in the first projection portion, a surface on the other side in the third direction has a concave shape, and a surface on the one side in the third direction is a half punched portion having a convex shape.
3. The actuator according to claim 1, wherein the coil holder is provided with a contact surface that makes contact with the first case first side plate portion from the other side in the third direction, and a substrate fixing portion that protrudes from the contact surface toward the one side of the first case first side plate portion in the third direction.
4. The actuator according to claim 1, wherein the coil holder comprises a first holder member provided with a first coil holding portion disposed on the one side of the coil in the third direction, and a first holder side plate portion that extends in the first direction from an end of the first coil holding portion on the one side in the third direction, and a second holder member provided with a second coil holding portion disposed on the other side of the coil in the third direction, and a second holder side plate portion that extends in the first direction from an end of the second coil holding portion on the other side in the third direction, the first holder member and the second holder member are fixed via a metallic first plate, which overlaps the first coil holding portion and the second coil holding portion from the one side in the first direction, and a metallic second plate, which overlaps the first coil holding portion and the second coil holding portion from the other side in the first direction, the first projection portion makes contact with the second holder side plate portion from the other side in the third direction, and the first holder side plate portion makes

contact with the first case first side plate portion from the other side in the third direction.

5. The actuator according to claim 4, wherein the movable body is provided with a yoke that holds the magnet, the yoke is provided with a first opposing portion that opposes the coil from the one side in the first direction, a second opposing portion that opposes the coil from the other side in the first direction, and a pair of connection portions that extend in the first direction on both sides of the coil in the second direction and connect the first opposing portion and the second opposing portion, the coil holder is provided with a pair of first holder projection portions that protrude toward the other side in the third direction from edges on both sides of the first holder side plate portion in the second direction, and a pair of second holder projection portions that protrude toward the one side in the third direction from edges on both sides of the second holder side plate portion in the second direction, an end portion of the first opposing portion and the second opposing portion on the one side in the third direction is disposed between the pair of first holder projection portions, and an end portion of the first opposing portion and the second opposing portion on the other side in the third direction is disposed between the pair of second holder projection portions.

6. The actuator according to claim 5, wherein the first case member is provided with a first case third side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on one side in the second direction, and a first case fourth side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on an other side in the second direction, and the pair of first holder projection portions and the pair of second holder projection portions are press-fitted between the first case third side plate portion and the first case fourth side plate portion.

7. The actuator according to claim 1, wherein the first case member is provided with a first case third side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on one side in the second direction, and a first case fourth side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on an other side in the second direction, the second case member is provided with a second case third side plate portion that extends toward the one side in the first direction from an end of the second end plate portion on the one side in the second direction, and a second case fourth side plate portion that extends toward the one side in the first direction from an end of the second end plate portion on the other side in the second direction, and the first case third side plate portion and the first case fourth side plate portion are press-fitted between the second case third side plate portion and the second case fourth side plate portion.

8. The actuator according to claim 7, wherein one of the second case third side plate portion and the first case third side plate portion is provided with a second projection portion that protrudes toward the other of the second case third side plate portion and the first case third side plate portion, and one of the second case fourth side plate portion and the first case fourth side plate portion is provided with a third projection portion that protrudes toward the other of the second case fourth side plate portion and the first case fourth side plate portion.

9. The actuator according to claim 8, wherein the second projection portion and the third projection portion are half punched portions.
